

Learning Personalized Agent for Real-Time Face-to-Face Interaction in VR

Xiaonuo Dongye
Beijing Institute of Technology

Dongdong Weng*
Beijing Institute of Technology

Haiyan Jiang
Beijing Institute of Technology

Pukun Chen
Beijing Institute of Technology

ABSTRACT

Interactive agents in virtual reality (VR) are anticipated to make decisions and provide feedback based on the user's inputs. Despite recent advancements in large language models (LLMs), employing LLMs in real-time face-to-face interactions decision-making, and delivering personalized feedback in VR remains challenging. To address this, our proposed system involves generating and labeling symbolic data, pre-training a real-time network, collecting personalized data, and fine-tuning the network. Utilizing inputs such as interaction distances, head orientations, and hand poses, the agents can provide personalized feedback. User experiments show significant advantages in both pragmatic and hedonic aspects over LLM-based agents, suggesting potential applications across diverse interactive domains.

Index Terms: Human-centered computing — Interaction design — Interaction design process and methods — Scenario-based design

1 INTRODUCTION

Interactive agents are computer-based entities designed to engage with users within virtual environments (VE), emphasizing quick and real-time responses. However, providing appropriate feedback in face-to-face interactions based on the user's input interaction data is a significant challenge for agents. The multitude of user interaction input modalities in virtual reality makes it challenging to collect and label decision-making datasets [1]. Recently, large language models have been employed as the primary decision hub for agents, as they are believed to possess some level of generalized intelligence [2]. However, LLM-based agents come with several limitations [3]. Interaction decision-making is challenging in real-time interaction as it relies on the frequent retrieval of extensive large language models. Additionally, large language models for decision-making necessitate additional approaches to align with the user's personalized interaction inputs.

In response to the challenges mentioned above, we aim to integrate the benefits of real-time interaction with the strengths of LLMs for generalized decision-making. This poster introduces a novel system designed to enable agents to provide real-time personalized interaction feedback in face-to-face interactions in VR. Our approach involves several key steps. Firstly, we generate random symbolic interaction data and utilize an LLM to label the generated data, creating a pre-train dataset. A small-scale decision-making network, capable of real-time operation, is trained with this dataset. Subsequently, a VE is constructed to facilitate face-to-face interactions between humans and agents. The agent utilizes the pre-trained network for decision-making while concurrently collecting personalized interaction data specific to each user during the interaction process. Finally, the collected personalized interaction data is used to fine-tune the network, enabling real-time decision-making that incorporates the user's personalized interaction inputs. Consequently, the agent becomes adept at comprehending users' inputs and delivering personalized feedback aligned with their inputs.

*e-mail: crgj@bit.edu.cn

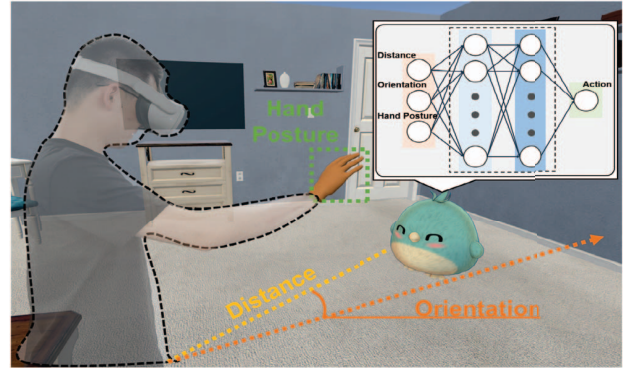


Figure 1: In face-to-face interactions within VR, interactive agents are required to provide appropriate action feedback by utilizing the user's real-time multimodal inputs. In our system, the agent incorporates personalized interaction decision-making using interaction distance, user head orientation, and user hand pose.

2 SYSTEM IMPLEMENTATION

Our system leverages the most prevalent user input modalities in consumer-grade head-mounted display (HMD) devices, which are interaction distance (the direct linear distance between the spatial coordinates of the HMD and the agent), head orientation (the angle formed between the rays originating from the center of the HMD and the line connecting the HMD to the agent), and hand pose (the hand rotational data tracked by the HMD) [4], as illustrated in Fig.1. These modalities serve as inputs for the agent, and based on these input modalities, the agent selects one of the symbolic actions it can perform as feedback.

2.1 Interaction Data Generation and Labeling

To enable real-time face-to-face interactions, we utilize a small-scale real-time neural network for the agent's decision-making process. This approach offers the advantage of reduced decision-making latency, enhancing user immersion compared to employing an LLM for decision-making. However, training such a network requires labeled interaction datasets, which are often lacking. To overcome this limitation, we leverage LLM's functionality. We let the LLM act as the agent, providing the symbolic inputs and letting it select action feedback. We generate random interaction distances up to 5m along with random rotation angles. Three distinct user interaction hand poses are considered: resting, palm facing the agent, and back facing the agent. We define a set of actions for the agent, including idle, rub forward, and jump up. To label the data, we first instruct OpenAI's "text-davinci-002" model regarding the environment, its task, the symbolized data inputs, and the symbolized action set outputs. Once the LLM comprehends its task, we adopt a multi-round dialogue approach and cyclically invoke the LLM and let it select action feedback multiple times. The actions selected by the LLM are stored as labels and paired with the randomly generated symbolic data.

2.2 Pre-Train Decision-Making Network for Agent

We utilize a three-classification network structure to facilitate the agent's interaction decision-making process. The decision-making network is a fully connected network featuring two hidden layers. We utilize the labeled generalized interaction data obtained from the

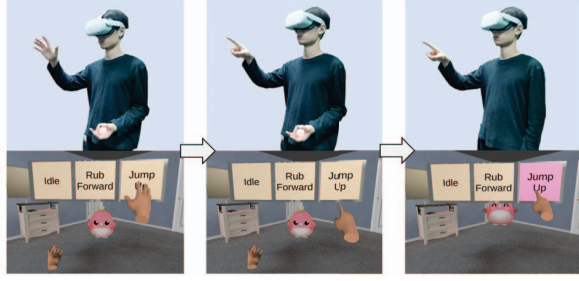


Figure 2: The user can pinch the thumb and middle finger of their left hand to engage the user interface (UI). The system records the interaction distance, head orientation, and right-hand pose at this moment. Subsequently, the user can select the corresponding action feedback from the UI by clicking a button. Once the user clicks this button, the agent changes action feedback and records it.

previous step. The training of this network is conducted employing the cross-entropy loss function, $L_{ce} = -\sum_{j=1}^3 t_j \log(p_j)$, where t_j is the truth label and p_j is the softmax probability for the j class.

2.3 Personalized Interaction Data Collection

To enable the agent to provide personalized interaction feedback, the most effective means of acquiring high-quality personalized interaction data is by directly obtaining it within the VR context [5]. We use the network trained in the previous step as a decision hub to drive the agent to interact. We set up an indoor scene of face-to-face interaction with a virtual pet using Meta Quest 2 for head tracking and hand tracking. In real-time interaction, the agent gives action feedback based on the user's position movement, head movement, and hand pose. During this interaction process, if the user perceives that the agent's feedback falls short of their expectations, they can engage the UI by pinching the thumb and middle finger of their left hand, as illustrated in Fig. 2. By clicking the UI buttons, users can select the agent's action feedback to align with their expectations. We record the user's button selections paired with the user's inputs for personalized decision-making fine-tuning.

2.4 Fine-Tune Network with Personalized Data

The personalized interaction input data collected for a specific user can be leveraged to fine-tune the general interaction decision-making network, creating a personalized interaction decision-making network. In our system, we only fine-tune the final fully connected layer. This choice was made to balance efficiency, computational cost, and the agent's ability to provide reasonably context-aware personalized feedback. Through multiple rounds of interaction, the agent becomes capable of providing personalized feedback based on the inputs from an individual user during the interaction process.

3 SYSTEM EVALUATION

We conducted a comparative experiment to assess whether our proposed system enhances the user's interaction experience compared to directly using an LLM to drive the agent for interaction. Twelve participants took part in the user studies, with six of them having prior experience with VR devices. Participants were informed that they would engage in face-to-face interactions with a virtual pet, utilizing positional walking, head movements, and right-hand postures for interaction. They were also provided information about the possible action feedback from the agent. All participants interacted with both the LLM-based agent and our system for 5 minutes in a randomized order. The agents in both cases had the same virtual environment, appearance, and feedback actions, differing only in the method of decision-making. After each interaction, participants were asked to score on the UEQ-S questionnaire.

We collected 12 results, as shown in Fig. 3. The data collected from each group conformed to a normal distribution. The mean

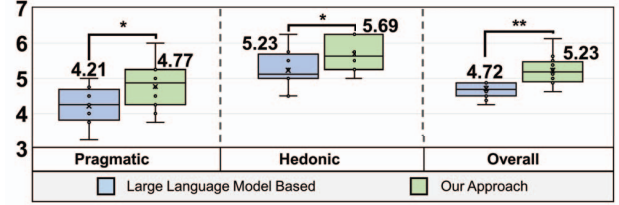


Figure 3: Subjective results of UEQ-S. “*” stands for significant difference ($p < 0.05$). “**” stands for significant difference ($p < 0.01$).

score for the LLM-based agent on the pragmatic scale was 4.21; on the hedonic scale, the LLM-based agent had a mean score of 5.23; and the mean score overall was 4.72. Our system received a mean score of 4.77 on the pragmatic scale; on the hedonic scale, the mean score was 5.69; and the mean score for overall was 5.23. In terms of the pragmatic scale, the paired t-test showed that there was a significant difference between the LLM-based agent and our system ($t_{11} = -2.66$, $p = 0.02$). Significant differences were also found in the hedonic aspect ($t_{11} = -2.99$, $p = 0.012$). In overall experience, there was a significant difference between our system and the LLM-based agent ($t_{11} = -4.54$, $p = 0.001$). The experimental results show that our system has a significant advantage over directly using LLM to drive the agent in both pragmatic and hedonic aspects, leading to an enhanced overall experience.

4 DISCUSSION AND FUTURE WORK

To address the challenges posed by LLMs in real-time personalized face-to-face interaction in VR, we present a system that involves generating and labeling general symbolic data, pre-training the network, collecting real interaction data, and fine-tuning the network. Continuous utilization of personalized interaction data enables ongoing refinement of the interaction decision-making network. Compared to an agent directly driven by an LLM in face-to-face VR interaction, our system has better ratings in user experience. There are several potential avenues for future research, such as exploring boundaries in systems with multiple input modalities and action sets, utilizing different network structures to handle continuous interaction inputs, and developing efficient data labeling approaches in VE. We hope that our work will inspire future researchers to design a wide variety of interactive agents.

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REFERENCES

- [1] Norina Gasteiger, Mehdi Hellou, and Ho Seok Ahn. Factors for personalization and localization to optimize human–robot interaction: A literature review. *International Journal of Social Robotics*, 15(4):689–701, 2023.
- [2] Sai Vemprala, Rogerio Bonatti, Arthur Buckner, and Ashish Kapoor. Chatgpt for robotics: Design principles and model abilities. *Microsoft Auton. Syst. Robot. Res.*, 2:20, 2023.
- [3] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*, 2023.
- [4] Xiaonuo Dongye, Dongdong Weng, Haiyan Jiang, and Lulu Feng. A modular haptic agent system with encountered-type active interaction. *Electronics*, 12(9):2069, 2023.
- [5] Mohamed El Beheiry, Sébastien Dautreigne, Clément Caporal, Cécilia Ostertag, Maxime Dahan, and Jean-Baptiste Masson. Virtual reality: beyond visualization. *Journal of molecular biology*, 431(7):1315–1321, 2019.